

Predicting Learner Disengagement on Lernnavi

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ABSTRACT

We study disengagement prediction on Lernnavi, a continuously available learning platform without fixed course-completion labels. We propose a two-stage framework adapted to this setting. Stage 1 separates early dabblers from continuing learners using aggregated activity from the first three weeks, while Stage 2 predicts sustained interruptions among continuing learners from 12-week behavioral histories. We compare Logistic Regression, LightGBM, and LSTM models using behavioral, performance, temporal, and demographic/contextual features. Results show that contextual and calendar variables are highly predictive: the best Stage 2 model reaches an AUC-ROC of 0.9125, while removing calendar features causes a major performance drop. In contrast, data augmentation brings little improvement. Fairness analysis further reveals substantial reliability differences across demographic groups, especially across cantons. Overall, our findings suggest that disengagement prediction on Lernnavi should be used as low-stakes decision support rather than as an automatic intervention system.

Keywords: learner disengagement; dropout prediction; educational data mining; LightGBM; LSTM; fairness

1. INTRODUCTION

Online learning platforms have broadened access to education, but they still face a persistent problem: many registered users disengage early. Predicting disengagement early enough to support learners has therefore become a central problem in educational data mining, with previous work often relying on clickstream features and sequence models [1, 5, 7].

Lernnavi, the Swiss high-school practice platform studied in this project, differs from standard MOOCs. It is a continuously available practice platform, with no video content, no fixed end date, and no enrollment cohort. This changes the prediction problem.

First, the usual MOOC label “did not complete the course” is not directly applicable, so disengagement must be defined

through inactivity gaps [5, 7].

Second, the user population contains many short-lived visitors: in our processed data, 78% of users are active in at most five distinct weeks.

We therefore separate disengagement into two related but different prediction tasks. Stage 1 focuses on early usage and predicts whether a learner comes back after the first three weeks of activity. This separates early dabblers from continuing learners. Stage 2 focuses only on continuing learners and predicts whether a learner is about to experience a long interruption in platform usage [7].

Our main research question is:

How can we predict learner disengagement from prior behavior and performance?

We divide this question into two subquestions:

1. *Can we distinguish early dabblers from continuing learners?*
2. *Can we predict sustained interruptions among continuing learners?*

We answer these questions with a two-stage pipeline using Lernnavi clickstream features, performance indicators, regularity statistics [8], and learner demographics. Stage 1 uses user-level features aggregated over weeks 0–2 and compares Logistic Regression with LightGBM. Stage 2 uses 12-week behavioral sequences and trains an LSTM model, with additional LightGBM experiments on flattened sequences for data augmentation analysis.

Because demographic features are included in the analysis, fairness evaluation is necessary. A model that performs well on average may still perform poorly for specific gender or canton groups, especially when some groups have small sample sizes. We therefore complement our predictive research questions with the following fairness questions:

1. *How does model performance vary across gender and canton groups?*
2. *Are there demographic groups for which the model is less reliable?*
3. *When demographic features improve overall predictive*

performance, do they also increase unfairness across groups?

Our contributions are threefold. First, we adapt disengagement prediction to the setting of a continuously available intelligent tutoring system without completion labels. Second, we compare tabular and sequential models under strong temporal seasonality constraints using a two-stage prediction framework. Third, we evaluate the fairness implications of demographic and contextual variables across gender and canton groups.

2. RELATED WORK

Predicting learner disengagement has become a major topic in educational data mining, particularly in MOOCs and intelligent tutoring systems. Prior work has shown that clickstream activity, temporal engagement patterns, and behavioral regularity are strong predictors of dropout risk [1, 5].

Several studies have specifically focused on sequential modeling approaches. Fu et al. [5] proposed deep learning architectures for MOOC dropout prediction using temporal learner activity, while Basnet et al. [1] compared deep learning and traditional machine learning approaches for engagement prediction. Similar to these works, our Stage 2 setup models learner behavior over time using weekly activity sequences.

Other work has emphasized the importance of behavioral regularity and wheel-spinning behavior. Beck and Gong [2] introduced the concept of wheel-spinning in intelligent tutoring systems, while Shirvani Boroujeni et al. [8] studied learner regularity as a predictor of educational outcomes. Our feature engineering strategy builds directly on these ideas.

Finally, fairness and demographic bias have become increasingly important in machine learning systems, especially when predictions may influence human decisions [6, 4]. Because disengagement prediction models may influence educational interventions, it is important to evaluate whether predictive performance differs across demographic groups. Our work extends prior disengagement prediction research by explicitly auditing fairness across gender and canton groups.

3. METHODOLOGY

3.1 Models and research questions

Our methodology follows the two research questions introduced above. For RQ1, which asks whether early dabblers can be distinguished from continuing learners, we use user-level tabular models trained on activity aggregated over the first three relative weeks. We compare Logistic Regression as an interpretable baseline with LightGBM, which is well suited to heterogeneous tabular features and nonlinear interactions.

For RQ2, which asks whether sustained interruptions can be predicted among continuing learners, we use weekly behavioral histories. We compare LightGBM on flattened 12-week histories with an LSTM model that preserves the temporal structure of the sequence. This comparison allows us to test whether explicit sequence modeling improves prediction beyond a strong tabular baseline.

3.2 Evaluation metrics

We evaluate predictive performance using AUC-ROC and F_1 -score. AUC-ROC measures how well the model ranks positive and negative examples across decision thresholds, while F_1 balances precision and recall at a chosen threshold. We use these two metrics because both stages are binary classification tasks and both types of classification error matter in an educational intervention setting: false positives may lead to unnecessary support, while false negatives may miss learners at risk of disengagement.

For Stage 1, the positive class corresponds to continuing learners, while early dabblers are the negative class. For Stage 2, the positive class corresponds to an interruption point. Because the interpretation of precision and recall depends on the positive class, we report AUC-ROC and F_1 together with the task definition.

3.3 Stage 1 task definition and features

At the first stage, we aim to distinguish early dabblers from learners who continue using Lernnavi after their first weeks of activity. We define **dabblers** as users who are active during the first three relative weeks but are not observed again afterward. This gives a user-level binary classification task: learners who come back after this initial window are labelled as continuing learners, while the others are labelled as early dabblers. The three-week cut-off is motivated by the empirical activity distribution: approximately 40% of users do not return after their first three weeks on the platform.

For Stage 1, we aggregate each user’s activity over the first three relative weeks. The baseline features describe early engagement and performance, such as activity volume, active days, sessions, transactions, opened documents, topic coverage, correctness rate, evaluation score, and response time. We then extend this baseline with session-duration statistics [3], retry and review behavior inspired by wheel-spinning analysis [2], and variability features measuring irregularity in activity time and response time [9]. Finally, we include demographic and contextual variables, such as gender, canton, class level, school type, and study-related indicators. Since these variables are not direct learning behaviors, we treat them separately from behavioral features. We use them to quantify their predictive contribution and to audit whether model reliability differs across groups, but we do not interpret them as actionable causes of disengagement.

3.4 Stage 2 task definition and features

In the second stage, we focus on learners who continue using Lernnavi after the early activity window. Instead of asking whether they leave immediately, we study whether their learning activity is interrupted later on. We define a sustained interruption as a period of at least four consecutive inactive weeks after a learner has already shown continued activity. Interruptions that overlap with July or August are excluded, because they are likely to correspond to summer holidays rather than meaningful disengagement.

3.4.1 Data preprocessing

For this task, there is a strong temporal dependence, so we could not rely on aggregated features as in the previous task. After an overall first preprocessing with missing values

treatment and outliers analysis, we opted for week-by-week data, i.e. we aggregated the activity data of the same week for each user. Thus our dataset has one row per user per week, with weeks indexed relative to each learner’s first active week. For each user and each week, we extract behavioral, temporal, performance, and engagement-related features.

- *user_id* : the id of the user
- *relative_week* : the index of the week, the first week of activity of the user is week 0.
- *n_events* : the number of events of the user on this row’s week.
- *n_active_days* : number of different days of activity for the user on this row’s week.
- *mean_hour* : average of the time of the day at which the user made the events during this row’s week.
- *n_click_events* : number of events of type ”click” during this week.
- *n_view_events* : number of events of type ”view” during this week.
- *n_sessions* : number of different sessions for the user and the week.
- *n_topics_events* : number of different topics that the user worked on this week.
- *n_transactions* : number of transactions of the user this week.
- *correct_rate* : rate of answers rated ’CORRECT’ among all the submissions from this user this week.
- *partial_rate* : rate of answers rated ’PARTIAL’ among all the submissions from this user this week.
- *mean_evaluation_score* : mean points given to the submissions from this user this week, according to the Lernnavi’s grading system.
- *avg_response_time* : average time taken by the user for the answers this week.
- *n_topics_transaction* : number of different topics in the transactions for this user this week.
- *n_documents* : number of different documents (questions) opened by the user this week.
- *activity_score* : aggregated score of activity from number of events and transactions.
- *day* : day of the year (1-365) of the beginning of the week
- *year* : year of the week.

We label each user-week as an **interruption point** if the following four weeks have an activity score of zero. Each model input is built from the previous 12 weeks of user activity, and the model predicts whether the learner is entering such an interruption.

3.4.2 Data Augmentation

This experiment is motivated by the imbalance observed in the original result-grade labels: **UNKNOWN** appears 733,752 times, **CORRECT** 668,991 times, **PARTIAL** 548,658 times, but **WRONG** only 183,358 times. However, after preprocessing and converting the task into the final binary interruption prediction problem, the training target is almost balanced. Therefore, augmentation is not expected to solve a strong class-imbalance issue, but rather to test whether synthetic variation can improve robustness.

All augmentation methods are applied only to the training set, while the validation and test sets are kept unchanged to avoid information leakage. We test four augmentation strategies. First, SMOTE generates synthetic samples by interpolating between nearby training examples. Second, Mixup creates new examples by interpolating pairs of training samples. Third, feature masking randomly hides a small subset of input features and replaces them with column medians. Finally, continuous noise adds small Gaussian perturbations only to continuous features.

3.4.3 Training and evaluation

For Stage 2, LightGBM is trained on flattened 12-week histories, while the LSTM preserves the sequential structure of the weekly activity data. The LSTM is trained with the Adam optimizer, a learning rate of 10^{-3} , and early stopping based on validation performance. We split the dataset by user to avoid having the same learner appear in both training and test sets: 20% of users are held out for testing, and 20% of the remaining users are used for validation.

We evaluate three complementary Stage 2 setups. First, we compare the main model families on the final interruption-prediction task. Second, because user activity shows strong seasonality, we run a calendar-feature ablation with LightGBM to isolate the effect of the *day* and *year* features. Third, we test whether data augmentation improves robustness on the same binary prediction task. We also compare models with and without demographic features to measure their predictive contribution and their effect on fairness.

3.4.4 Trajectory clustering

In addition to the supervised prediction experiments, we conduct an exploratory trajectory-clustering analysis for Stage 2. We standardize weekly behavioral features and group continuing learners using spectral clustering. The clustering uses activity, regularity, performance, and topic-breadth signals over the learner history. This analysis is not used for model training; it is used only to interpret common behavioral profiles and compare interruption rates across school types.

3.5 Fairness evaluation

After evaluating predictive performance, we audit whether the models behave similarly across demographic groups. This is important because demographic and contextual variables improve performance, but they may also make predictions less reliable for specific groups. We focus on two classification fairness criteria: equalized odds and predictive value parity.

- Equalized odds : for any label or demographic feature, the model predicts equally well for each value of this

demographic feature. More formally, for a demographic variable A , the true target label $T \in \{0, 1\}$ and the model's prediction $\hat{Y}$, we want :

$$p(\hat{Y} = 1|T = 1, A = a) = p(\hat{Y} = 1|T = 1, A = a')$$

$$p(\hat{Y} = 1|T = 0, A = a) = p(\hat{Y} = 1|T = 0, A = a')$$

for any a, a' possible values of A .

- Predictive value parity : Given a model prediction for a sample, the probability that this sample belongs in fact to the predicted class should be the same across all values of that attribute. Again, for a demographic variable A , the true dropout label $T \in \{0, 1\}$ and the model's prediction $\hat{Y}$, we want :

$$p(T = 1|\hat{Y} = 1, A = a) = p(T = 1|\hat{Y} = 1, A = a')$$

$$p(T = 0|\hat{Y} = 0, A = a) = p(T = 0|\hat{Y} = 0, A = a')$$

for any a, a' possible values A .

We need to test these equalities for our demographic features. For each metric, we compute the maximum absolute difference between groups for the probabilities of interest. More formally, for equalized odds we report :

$$\max_{a, a'} |p(\hat{Y} = 1|T = 1, A = a) - p(\hat{Y} = 1|T = 1, A = a')|$$

and

$$\max_{a, a'} |p(\hat{Y} = 1|T = 0, A = a) - p(\hat{Y} = 1|T = 0, A = a')|$$

for some demographic feature A . And for predictive value parity, we report :

$$\max_{a, a'} |p(T = 1|\hat{Y} = 1, A = a) - p(T = 1|\hat{Y} = 1, A = a')|$$

$$\max_{a, a'} |p(T = 0|\hat{Y} = 0, A = a) - p(T = 0|\hat{Y} = 0, A = a')|$$

for some demographic feature A .

We computed these metrics for two different demographic features : gender and canton. We do not combine gender and canton into intersectional groups, because some cantons already contain very few samples and the resulting strata would be too small or empty. Table 1 reports the repartition of users across values for the gender feature.

Table 1: Sample distribution by gender.

Gender	Number of samples
Female	15342
Male	10364
Other	1006
Unknown	4217

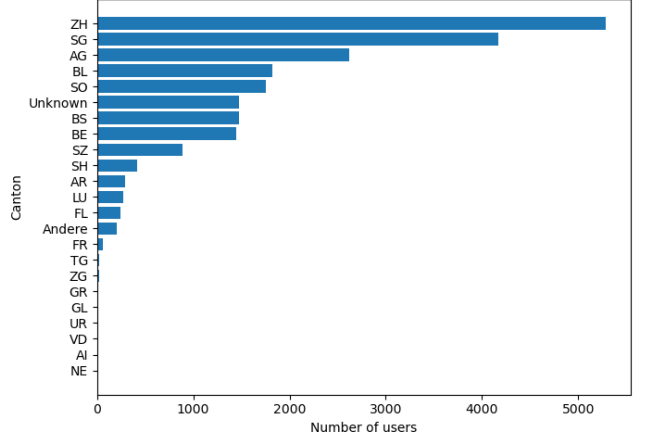


Figure 1: User distribution by canton. The imbalance motivates caution when interpreting canton-level fairness gaps.

Figure 1 shows a strong imbalance in canton representation: a few cantons contain many users, while several others contain very small groups. This imbalance is important for the fairness analysis, because group-level metrics can become unstable when computed on only a few samples. We therefore interpret canton-level fairness gaps cautiously.

We performed this work both for stage 1 and stage 2, results can be seen in the Results section.

4. RESULTS

4.1 Stage 1: Early dabbler prediction

4.1.1 Predictive performance

Table 2 reports the cumulative feature-set ablation for Stage 1. The best model is LightGBM with the full feature set, reaching an AUC-ROC of 0.7711 and an F_1 -score of 0.7909. It performs consistently better than the Logistic Regression baseline, especially after adding demographic and contextual variables.

Table 2: Stage 1 feature-set ablation.

Features	LightGBM		Logistic Regression	
	AUC-ROC	F_1	AUC-ROC	F_1
Baseline	0.6359	0.7545	0.6200	0.7560
+Additional	0.6480	0.7565	0.6243	0.7560
+Demographic	0.7711	0.7909	0.7087	0.7624

4.1.2 Feature importance and interpretation

The additional behavioral features bring only a small improvement: LightGBM AUC-ROC increases from 0.6359 to 0.6480, and F_1 increases from 0.7545 to 0.7565. Most of the gain comes from the demographic and contextual feature block, which increases LightGBM AUC-ROC from 0.6480 to 0.7711 and F_1 from 0.7565 to 0.7909. Logistic Regression shows the same pattern, with AUC-ROC increasing from 0.6243 to 0.7087 after adding demographic and contextual features.

This suggests that early continuation on Lernnavi is not explained only by individual activity during the first weeks.

It is also associated with contextual factors such as canton, class level, school type, or the way Lernnavi is integrated into the learner’s educational environment. This result motivates the fairness analysis, because demographic and contextual variables improve predictive performance but may also create unequal reliability across groups.

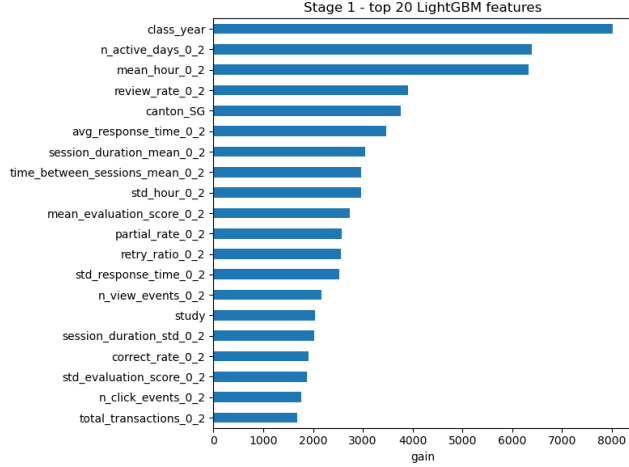


Figure 2: Top 20 LightGBM feature importances for Stage 1.

Figure 2 confirms that the Stage 1 model relies on both early behavior and demographic/contextual information.

4.2 Stage 2: Sustained interruption prediction

4.2.1 Predictive performance

The Stage 2 result tables correspond to different experimental setups. Table 3 reports the main model comparison, Table 4 reports the calendar-feature ablation, and Table 5 reports the data augmentation experiments. Because these experiments use different feature sets or training procedures, their baseline values are close but not identical.

Table 3 compares the main Stage 2 models. LGBM denotes LightGBM. LightGBM is trained on flattened 12-week sequences, while the LSTM keeps the temporal structure of the input. With weekly behavioral features only, LightGBM slightly outperforms the LSTM, reaching an AUC-ROC of 0.8929 and an F_1 -score of 0.8313, compared with 0.8837 and 0.8226 for the LSTM.

The best overall model is the LSTM with demographic features, reaching an AUC-ROC of 0.9125 and an F_1 -score of 0.8397. This confirms that demographic and contextual variables improve predictive performance in Stage 2, but also reinforces the need for the fairness analysis.

Table 3: Stage 2 predictive performance.

Model	AUC	F_1	Prec.	Rec.
LGBM weekly	0.8929	0.8313	0.7469	0.9372
LGBM no-summer	0.8417	0.8317	0.7485	0.9356
LGBM time-only	0.8291	0.7982	0.6719	0.9830
LSTM weekly	0.8837	0.8226	0.7330	0.9372
LSTM no-summer	0.8376	0.8291	0.7514	0.9247
LSTM time-only	0.8183	0.7968	0.6656	0.9925
LSTM + demo.	0.9125	0.8397	0.7669	0.9278

4.2.2 Seasonality and feature ablation

Figure 3 shows that Lernnavi activity varies strongly over calendar time, with clear low-activity periods and large activity peaks. Table 4 confirms that this calendar structure is highly predictive of sustained interruptions. Removing the year has only a small effect, but removing the day-of-year feature reduces AUC-ROC from 0.8924 to 0.6993 and F_1 from 0.8348 to 0.6843. This large drop shows that the model partly learns school-calendar structure rather than only individual learner behavior. This improves predictive performance, but it also limits how directly the predictions can be interpreted as learner-level disengagement risk.

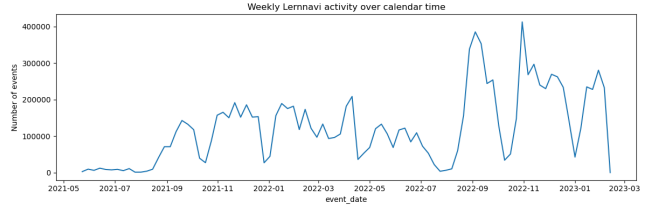


Figure 3: Weekly Lernnavi activity over calendar time, showing strong seasonal activity patterns and extended low-activity periods during holidays.

Table 4: Stage 2 calendar-feature ablation for LightGBM.

Features	AUC-ROC	F_1
Baseline	0.8924	0.8348
Removing year	0.8871	0.8286
Removing day	0.6993	0.6843
Removing year and day	0.5988	0.6797

4.2.3 Data augmentation

Table 5 shows that data augmentation does not meaningfully improve performance. The best augmented model, continuous noise, reaches an F_1 -score of 0.8313 compared to 0.8304 for the baseline, which is negligible. SMOTE even performs slightly worse. This suggests that the final binary dataset is already sufficiently large and balanced, and that future improvements should focus on temporal modeling and feature engineering rather than synthetic data generation.

Table 5: Stage 2 data augmentation results with LightGBM.

Method	AUC	Acc.	Prec.	Rec.	F_1
Cont. noise	0.8911	0.8157	0.7484	0.9349	0.8313
Masking	0.8903	0.8156	0.7495	0.9316	0.8307
Baseline	0.8906	0.8153	0.7492	0.9314	0.8304
Mixup	0.8898	0.8148	0.7488	0.9309	0.8300
SMOTE	0.8882	0.8125	0.7476	0.9268	0.8276

4.2.4 Trajectory clustering

To complement the predictive results, we examine whether the Stage 2 learner histories form interpretable behavioral profiles.

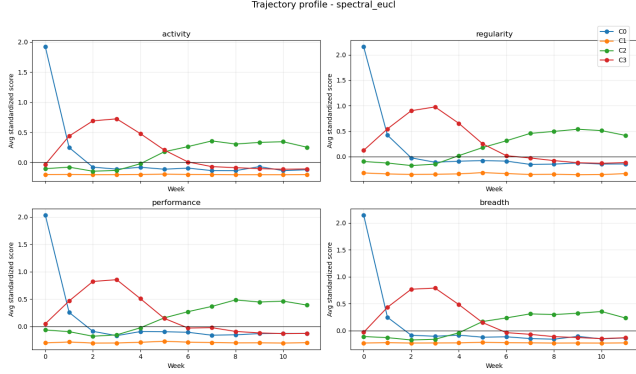


Figure 4: Average standardized behavioral trajectories for the four spectral-clustering profiles.

Figure 4 shows that continuing learners can be grouped into distinct trajectory profiles. Some clusters show early high activity followed by a rapid decrease, while others show more stable or delayed activity patterns. This supports the idea that sustained interruptions are related not only to overall activity volume, but also to the temporal shape of learner behavior.

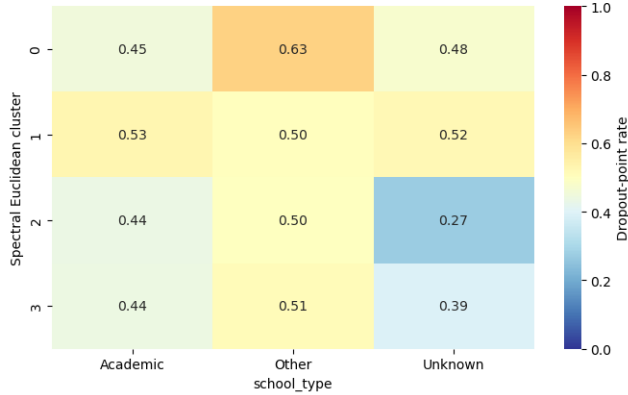


Figure 5: Interruption-point rate by behavioral cluster and school type.

Figure 5 shows that interruption rates differ by school type within the same behavioral cluster. This suggests that contextual variables can still matter even after grouping learners by similar activity trajectories.

4.3 Fairness results

Figures 6 and 7 report the maximum group-level gaps for the fairness metrics.

For Stage 1, gender gaps are relatively limited for equalized odds, with a TPR gap of 0.0689 and an FPR gap of 0.0877,

while the largest gender difference is observed for NPV with a gap of 0.1953. Canton-level gaps are much larger: the TPR gap reaches 0.6667, the FPR gap 0.8182, and the NPV gap 1.0000.

For Stage 2, adding demographic features has mixed fairness effects. For gender, the FPR and PPV gaps decrease slightly, but the TPR gap increases from 0.0227 to 0.1009. For canton, adding demographic features reduces the PPV gap from 0.9332 to 0.2356 and the FPR gap from 0.5000 to 0.3571, but increases the TPR gap from 0.2777 to 0.3767 and the NPV gap from 0.3125 to 0.4375.

Overall, the fairness audit shows that demographic and contextual variables improve predictive performance but do not automatically make predictions fairer. Gender gaps remain moderate, while canton-level gaps are larger and should be interpreted cautiously because several canton groups contain few users. This supports using demographic variables mainly for auditing and calibration, not as standalone explanations or intervention triggers.

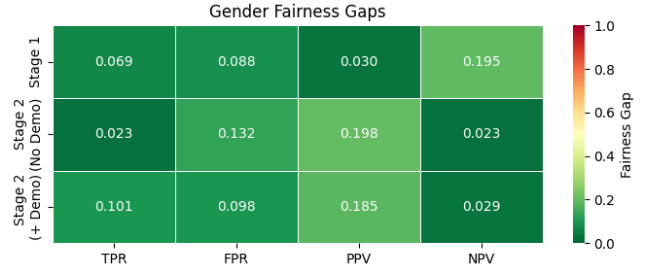


Figure 6: Maximum fairness gaps for gender

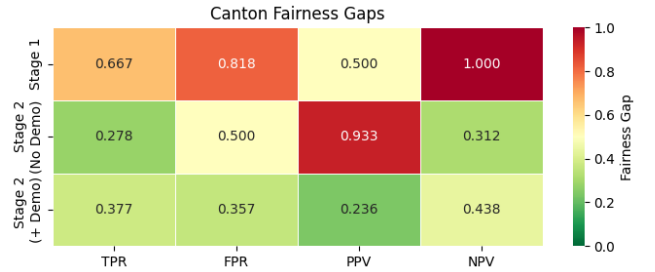


Figure 7: Maximum fairness gaps for canton

5. DISCUSSION

The two prediction tasks lead to different conclusions. Early dabbblers can be distinguished from continuing learners, but Stage 1 performance remains moderate and depends strongly on contextual variables. In contrast, sustained interruptions among continuing learners are predicted more accurately, especially when temporal and demographic information is included.

These findings are broadly consistent with prior MOOC disengagement research, which also reports strong predictive value from temporal engagement signals and sequential activity patterns [1, 5]. However, unlike traditional MOOCs,

Lernnavi operates as a continuously available intelligent tutoring system without fixed course completion targets. As a result, calendar effects and institutional context play a substantially larger role in disengagement prediction.

In Stage 1, additional behavioral features bring only small gains over the baseline activity and performance features. Most of the improvement comes from demographic and contextual variables. This suggests that early continuation on Lernnavi is not determined only by what learners do during their first weeks, but also by the educational context in which the platform is used. Variables such as canton, class level, school type, or study-related indicators may capture differences in how Lernnavi is integrated into school practice.

This result is useful but also risky. Contextual variables improve predictive performance, but they are not directly actionable learning behaviors. If the model relies too heavily on canton or school-related information, it may learn institutional usage patterns rather than individual disengagement. The fairness results confirm this concern: gender gaps are moderate overall, while canton-level gaps are much larger and more unstable, partly because some canton groups contain very few users. Demographic and contextual variables should therefore be used mainly for auditing and calibration, not as standalone explanations or triggers for intervention.

Stage 2 gives stronger predictive results than Stage 1. This is expected because the model observes a longer 12-week history and focuses only on learners who have already shown continued engagement. The LSTM with demographic features gives the best overall performance, but LightGBM also performs strongly on flattened weekly histories. This shows that the weekly activity sequence contains a large amount of predictive signal, even when the temporal structure is simplified.

A central Stage 2 result is the importance of calendar effects. Removing the day-of-year feature causes a large performance drop, which shows that sustained interruptions are strongly seasonal. The model is therefore not only learning individual learner trajectories, but also school-calendar rhythm. This matters for deployment: an inactivity period close to a holiday or exam period should not be interpreted in the same way as an inactivity period during a regular school week.

The clustering analysis supports the same interpretation. Continuing learners follow different behavioral trajectories in terms of activity, regularity, performance, and topic breadth. Moreover, interruption rates vary by school type even within similar behavioral clusters. This suggests that disengagement on Lernnavi is both behavioral and contextual: learner activity matters, but so does the educational setting in which the activity occurs.

Finally, the data augmentation experiments bring almost no improvement over the baseline. The best augmented model only slightly improves the F_1 -score, and the difference is negligible. Since the final binary interruption dataset is already large and nearly balanced, future improvements are more likely to come from better temporal validation, better treatment of calendar effects, and more interpretable behavioral features than from synthetic data generation.

6. LIMITATIONS

This work has several limitations. First, disengagement is defined only through platform inactivity, which does not necessarily reflect a learner losing motivation. A learner may stop using Lernnavi because of holidays, exams, teacher decisions, illness, or because they already achieved their learning objective. In addition, the strong predictive power of calendar variables shows that the models partly learn school-calendar structure rather than only individual disengagement risk. Although summer interruptions are excluded from the labels, other institutional and seasonal effects remain present in the data.

A second limitation concerns generalization and fairness. The evaluation split avoids user-level leakage, but it does not fully test whether the models generalize across future school years or changing educational settings. Moreover, demographic and contextual variables improve predictive performance while creating unequal reliability across groups, especially for small canton populations where fairness metrics become unstable. Future work should therefore focus on temporal validation across school years, more interpretable behavioral representations, and controlled intervention experiments measuring whether model-driven support actually improves learner outcomes in practice.

7. CONCLUSION

This project studied learner disengagement on Lernnavi through a two-stage prediction pipeline combining tabular and sequential models. Early dabblers can be distinguished from continuing learners with moderate performance, while sustained interruptions among continuing learners can be predicted more accurately using temporal behavioral histories. The results show that disengagement on Lernnavi is shaped not only by learner behavior, but also by contextual and calendar-related factors. Although demographic and contextual variables improve predictive performance, they also create fairness concerns and unequal reliability across groups. Overall, the models are best viewed as low-stakes early-warning systems that support teachers and platform developers rather than automatic intervention tools.

8. SUGGESTIONS TO STARTUP

Based on our results, Lernnavi should use disengagement prediction as a support tool rather than as an automatic intervention system. A reasonable first deployment would be a teacher-facing dashboard that highlights interpretable behavioral risk signals, such as decreasing activity, irregular study time, reduced topic breadth, or long inactivity risk. The dashboard should avoid deterministic labels such as “dropout student” and should keep the final intervention decision with teachers or platform staff. Since time-related and regularity features are important, Lernnavi could also test personalized study-time reminders through an A/B experiment, comparing standard reminders with reminders sent around each learner’s usual activity time. The evaluation should measure not only return rate, but also session quality, response-time stability, activity consistency, and topic breadth.

Future versions of the platform should collect richer contextual and intervention-related signals to make disengagement analysis more reliable. Useful features would include teacher assignment metadata, notification interaction logs, learner goal-setting information, school-calendar events, exam periods, and curriculum alignment indicators. These features would help distinguish voluntary inactivity from genuine disengagement and would reduce the risk that the model simply learns canton-, school-, or calendar-specific usage patterns. Demographic and contextual variables should mainly be used for auditing and calibration, not as standalone reasons for intervention.

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